

Multi-source land-use emissions reveal rising airborne fraction

J. Eduardo Vera-Valdes
Aalborg University, CoRE
eduardo@math.aau.dk

2026-05-27

Abstract The airborne fraction is the share of anthropogenic carbon dioxide emissions that remains in the atmosphere and is a key indicator of carbon-cycle response under continued emissions. Whether this share is rising remains debated because inference is sensitive to uncertainty in land-use and land-cover change (LULC) emissions. Here we use all available LULC measurement series from Global Carbon Budget 2025 data and estimate trends with a mixed-effects model with random intercepts and slopes by LULC series. We find that airborne fraction increased over 1959-2024, and that this conclusion is robust to excluding the final year and to a measurement-error-aware weighted least squares robustness specification. These results strengthen evidence that an increasing share of emitted carbon dioxide is accumulating in the atmosphere rather than being taken up by land and ocean sinks, with implications for carbon-budget assessment and near-term mitigation requirements.

1 Background

2 Whether the airborne fraction (AF) is increasing or approximately constant remains contested¹⁻⁷. In
3 its classical form, AF is a yearly ratio of atmospheric growth to total anthropogenic emissions,
4 computed as the sum of fossil fuel emissions and land-use and land-cover change emissions as

$$AF_t = \frac{G_t}{FF_t + LULC_t}, \quad (1)$$

5 where G_t is the annual atmospheric CO_2 growth, FF_t is fossil fuel emissions excluding carbonation,
6 and $LULC_t$ is land-use and land-cover change emissions. AF is a key carbon-cycle diagnostic, with
7 implications for carbon-cycle feedbacks and near-term mitigation planning^{1,2,8}.

8 A persistent concern is that AF inference depends on the treatment of land-use and land-cover
9 change (LULC) emissions, which are uncertain and model-dependent in annual carbon-budget
10 accounting. The Global Carbon Budget (GCB) 2025⁸ provides one column of LULC emissions as the

average of three bookkeeping models (BLUE⁹, OSCAR¹⁰, LUCE¹¹), but a broader set of model-based LULC alternatives can be constructed from the same source. Estimates that do not incorporate this multi-source LULC information can be underpowered and inference on trend direction becomes less reliable.

Here we address that issue with a statistical design that uses all repeated LULC measurements in a mixed-effects trend framework to obtain more reliable inference. The main specification estimates AF trends from a panel of yearly AF values constructed from each LULC measurement series, with random intercepts and random slopes capturing between-series heterogeneity and inducing within-series dependence over time. As a robustness exercise, we also estimate a two-stage measurement-error weighted least squares (WLS) model that propagates denominator uncertainty from repeated LULC measurements into annual AF uncertainty and uses heteroskedasticity-and-autocorrelation-consistent (HAC) inference^{12–14}.

Across specifications that incorporate the multi-source LULC information, we find robust evidence that AF has increased over 1959-2024, and that this conclusion is not driven by the final observation (2024), which shows a large AF value. By contrast, ordinary least squares (OLS) specifications that do not incorporate denominator measurement information provide weaker evidence of a positive trend and are more sensitive to endpoint exclusion. Taken together, the results clarify why AF trend significance has been elusive in the literature and strengthen evidence that a growing share of emitted carbon dioxide is accumulating in the atmosphere.

30 **Data**

We use annual Global Carbon Budget 2025 data for 1959-2024, with atmospheric growth G_t from NOAA/ESRL global concentration trends¹⁵, fossil emissions excluding carbonation FF_t from the Global Carbon Project fossil dataset⁸, and a panel of 69 LULC measurements per year: BLUE⁹, OSCAR¹⁰, LUCE¹¹, and peat-augmented^{16–18} process-based land-model combinations drawn from the GCB model ensemble^{19–39} (see Methods). The data is shown in Figure 1.

Two LULC means can be constructed from the panel. First, the GCB LULC mean, defined as the mean of the three bookkeeping models (BLUE, OSCAR, and LUCE), corresponds to the LULC column used in the Global Carbon Budget. Second, the all-LULC mean, defined as the cross-series mean across all 69 LULC measurements. The literature has typically used the GCB LULC mean as the denominator in AF trend estimation, but we show that incorporating the full set of LULC measurements increases the precision of trend estimates and provides more robust inference on trend direction.

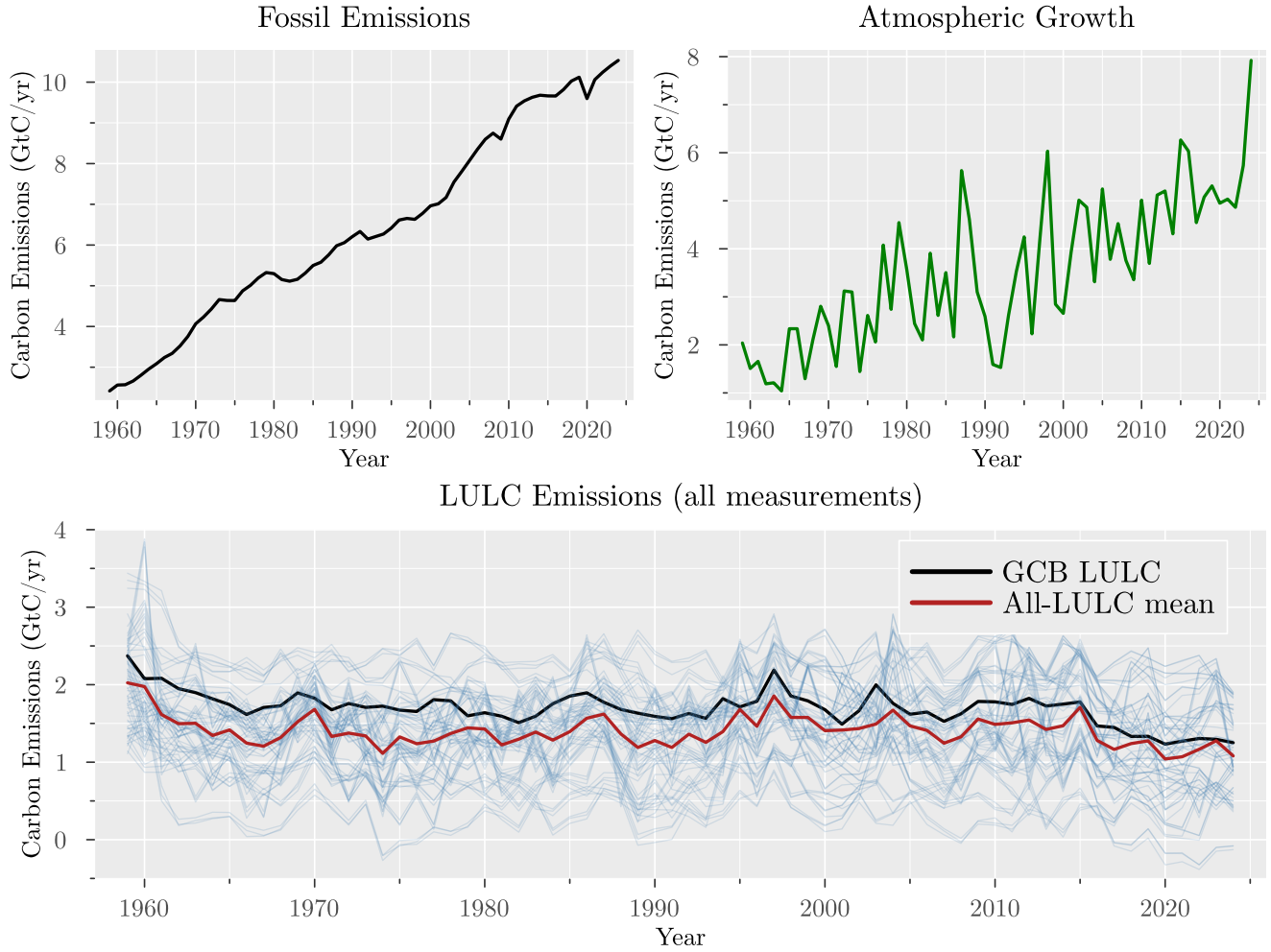


Figure 1: Global Carbon Budget 2025 data for fossil emissions, atmospheric growth, and LULC

emissions (1959-2024). For the LULC panel, we show all extracted and derived LULC measurements together with the GCB LULC column and the all-LULC cross-series mean.

Identification strategy

The empirical question is whether AF has a positive linear trend. The key design choice is how to use the full panel of repeated LULC measurements while accounting for within-series dependence and cross-series heterogeneity. OLS on a collapsed annual series discards panel information and hence provides less reliable inference, see results in Table 2. Our preferred specification is a mixed-effects model with random intercepts and random slopes by LULC series, allowing correlation across years within each series and capturing unobserved heterogeneity in LULC measurements^{40,41} (see Methods). For each LULC measurement series, we construct a series of AF estimates by year (Equation 1), and the model estimates a common time trend across all series while allowing for series-specific intercepts and slopes. Formally, the model can be written as

$$AF_{t,j} = \alpha_j + \beta_j t + \varepsilon_{t,j},$$

56 where

$$\alpha_j \sim N(\mu_\alpha, \sigma_\alpha^2), \quad \beta_j \sim N(\mu_\beta, \sigma_\beta^2), \quad \varepsilon_{t,j} \sim N(0, \sigma_\varepsilon^2).$$

57 Here μ_α and μ_β are the fixed effects (overall intercept and slope), while σ_α^2 and σ_β^2 capture the
58 variability in intercepts and slopes across LULC measurement series. The error term $\varepsilon_{t,j}$ captures
59 idiosyncratic noise. The key parameter of interest is μ_β , which captures the average trend across all
60 LULC measurement series. Incorporating the full panel of LULC measurements in this way allows us
61 to obtain more precise and robust estimates of the AF trend, while accounting for measurement
62 uncertainty and heterogeneity across LULC series.

63 Results

64 Main estimates

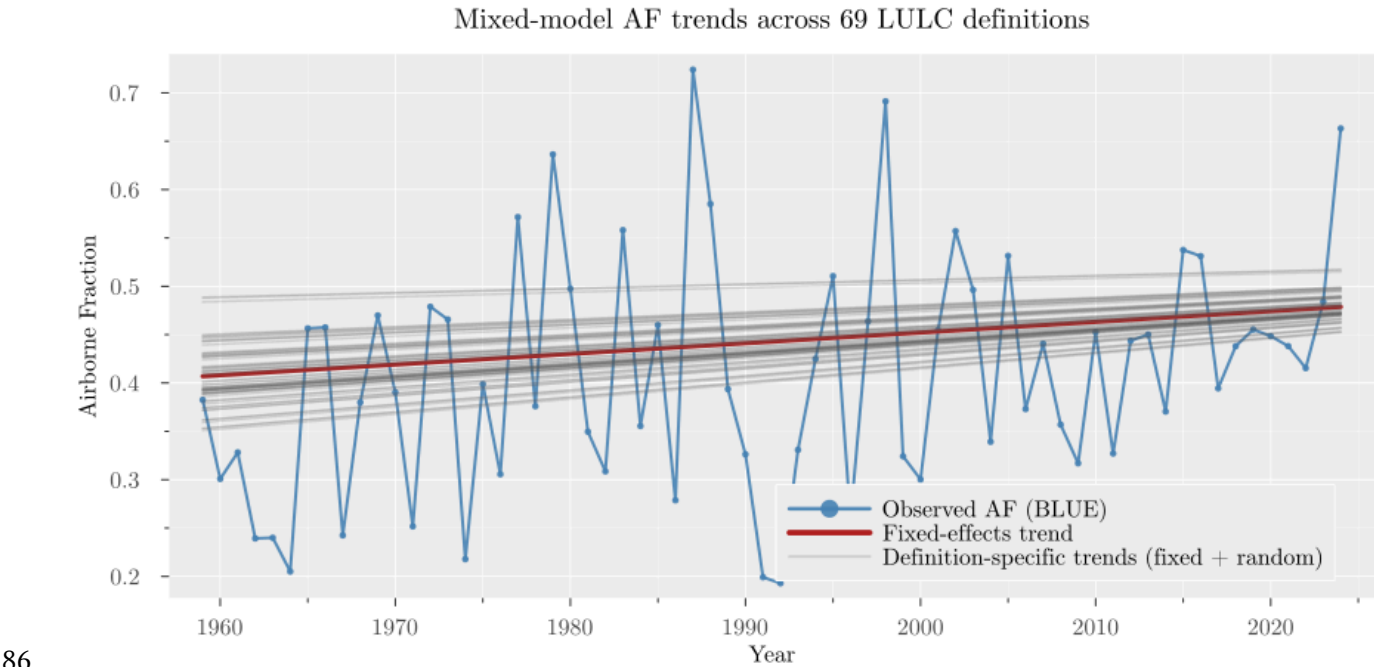
65 The main results from estimating the mixed-effects model over the full sample (1959-2024) are shown
66 in Table 1, and the associated trend lines are shown in Figure 2a.

67 Table 1: Mixed-effects model with random effects for each LULC measurement series.

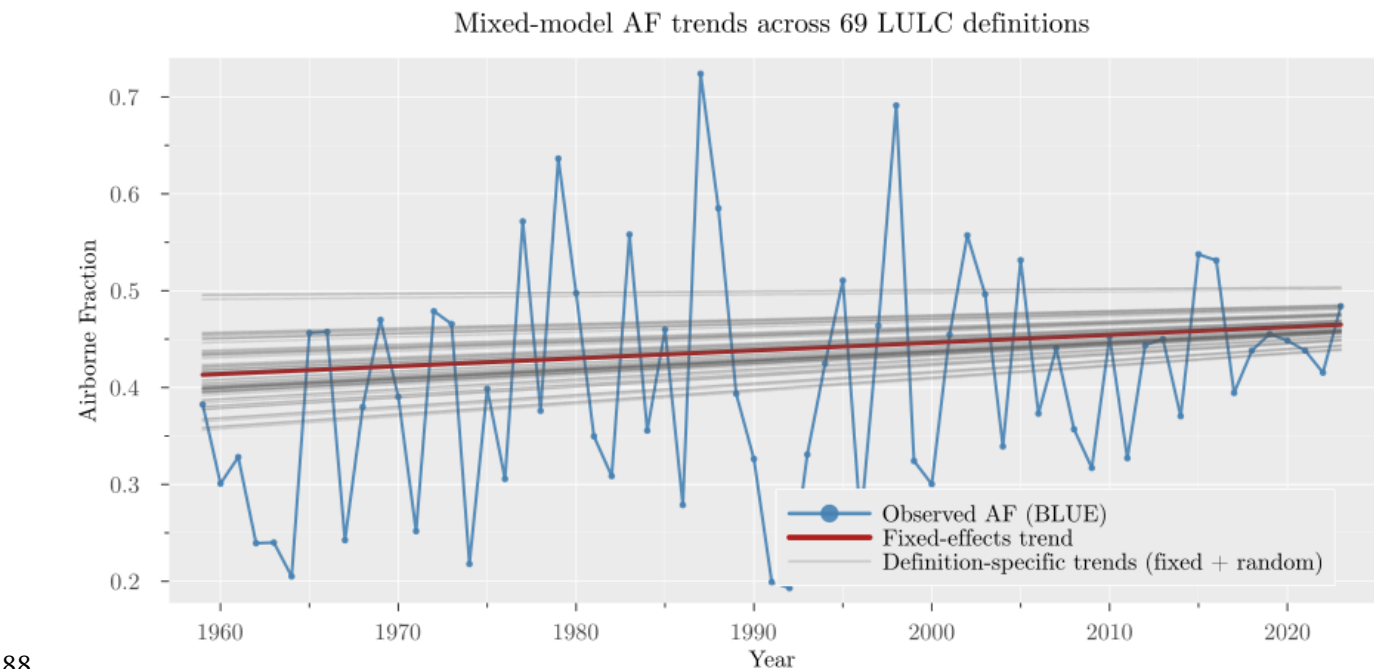
68 Parameter	Intercept (full)	Slope (full)	Intercept (up to 69 2023)	Slope (up to 2023)
70 Estimate	0.406959	0.001103	0.413260	0.000808
71 Standard	0.005573	0.000102	0.005586	0.000103
72 error				
73 p-value	0.000000	0.000000	0.000000	0.000000
74 R-squared	0.008631	0.008631	0.004518	0.004518
75 (marginal)				
76 R-squared	0.700967	0.700967	0.708173	0.708173
77 (conditional)				

78 The mixed-effects slope is positive and statistically significant (p-value < 1E-5), indicating strong
79 evidence of an increasing AF trend across LULC measurement series. The estimated intercept is
80 around 0.41 and the slope is around 0.0011 per year, implying an increase of about 0.07 in AF over
81 1959-2024. These estimates are comparable with the approximately 0.45 AF level reported in the
82 literature^{2,3,6,7,42,43}, while identifying a positive long-run trend when all LULC information is
83 incorporated. The conditional R-squared values indicate that most explained variation is captured by

84 between-series structure rather than by the fixed linear time trend (marginal versus conditional R-
85 squared).



(a) AF series with mixed-effects model trend (full sample, 1959-2024).



(b) AF series with mixed-effects model trend (sample ending in 2023).

Figure 2:

Figure 2a shows the AF series for each LULC measurement together with the overall mixed-effects trend line. The figure shows a clear positive trend across the panel of AF series, with the mixed-effects model capturing this common signal while allowing for series-specific heterogeneity.

Endpoint and model specification robustness

AF shows a large value in the final year (2024), so we re-estimate the mixed-effects model on the subsample ending in 2023 to check whether the result is driven by that observation. Results are shown in the last two columns of Table 1 and Figure 2b. The slope remains positive and statistically significant (p-value < 1E-5), indicating that the increasing AF trend is not driven by the final observation. The slope is smaller (0.000808 per year versus 0.001103 per year), consistent with a contribution from the 2024 large value, but the direction and inference are unchanged.

As an additional robustness check, we estimate a delta-method uncertainty proxy for each annual AF estimate using the cross-measurement dispersion of the full 69-series LULC panel, and use these proxies in WLS with HAC inference^{12–14,44}. Two WLS specifications are shown in Table 2: one using the GCB LULC mean as the denominator and one using the all-LULC mean. In both specifications, weights are constructed from the all-LULC panel dispersion so that the weighting scheme is held fixed across denominator definitions. The WLS results are consistent with the mixed-effects results and show a positive, statistically significant trend. Note that the slopes estimated from WLS are steeper than those from OLS or the mixed-effects model, which is consistent with the fact that WLS assigns less weight to years with higher denominator uncertainty, which tend to be the earlier years of the sample.

Table 2: Full-sample trend comparison across OLS and WLS specifications.

	Trend, full sample	OLS (GCB)	OLS (all)	WLS (GCB)	WLS (all)
Estimate		0.00158	0.00117	0.00265	0.00230
Standard error		0.00077	0.00079	0.00002	0.00002
HAC standard error		0.00062	0.00064	0.00065	0.00066
p-value		0.04021	0.13891	0.00000	0.00000
HAC p-value		0.01112	0.06846	0.00004	0.00050
R-squared		0.06171	0.03309	0.14899	0.11107

By contrast, OLS specifications that do not incorporate denominator measurement information show weaker evidence of a positive trend and greater endpoint sensitivity. In the full sample, OLS using GCB LULC is significant under both conventional and HAC inference (p-values 0.04021 and 0.01112),

122 whereas OLS using the all-LULC mean is not significant under either method (p-values 0.13891 and
123 0.06846).

124 In the sample ending in 2023 (reported in the Supplementary Information), WLS remains significant
125 under both mean definitions (HAC p-values 0.00005 and 0.00080), whereas OLS only remains
126 significant when using the GCB LULC mean (HAC p-values 0.02495 and 0.14322).

127 This sensitivity to denominator construction and endpoint exclusion highlights the value of
128 incorporating multi-source LULC information in AF trend estimation.

129 **Discussion and conclusion**

130 Using a framework that incorporates all available LULC information from Global Carbon Budget
131 2025, we find robust evidence that AF increased from 1959 to 2024. Methodologically, incorporating
132 multi-source uncertainty materially changes inference relative to plain OLS. Endpoint tests show that
133 this conclusion is not driven by the large value in 2024.

134 A rising AF implies that a larger share of emitted CO_2 remains in the atmosphere over policy-
135 relevant horizons, tightening near-term mitigation requirements for a given temperature
136 objective^{1,2,8}. The increasing AF signal is consistent with broader evidence that the climate system is
137 out of energy balance and has recently shown elevated warming and heating rates^{45–48}; these findings
138 should be interpreted in that wider risk context.

139 **Methods**

140 Our primary estimator is a linear mixed-effects trend model with random intercepts and random
141 slopes for each LULC measurement series^{40,41,49,50}. As a robustness check, we also estimate a
142 measurement-error-aware airborne fraction variance using repeated yearly denominator
143 measurements, followed by WLS trend estimation with HAC inference^{12–14,44,51}.

144 **Construction of the LULC measurement panel**

145 The repeated LULC measurements are built from the Global Carbon Budget 2025 dataset. We first
146 extract the three bookkeeping series: BLUE⁹, OSCAR¹⁰, LUCE¹¹. Then, for each process-based land-
147 model LULC series in the GCB ensemble that does not already include peat emissions^{19–39}, we add the
148 corresponding peat component to make it comparable to the bookkeeping models, which include
149 peat emissions by construction. Specifically, for each of the 22 process-based series we create three
150 peat-augmented variants by adding FAO_peat, LPX_Bern_peat, and ORCHIDEE_peat^{16–18}. This

151 produces 66 derived series, and together with BLUE/OSCAR/LUCE gives a panel of 69 yearly LULC
 152 measurements.

153 **Mixed-effects model**

154 The primary estimator is a linear mixed-effects trend model fitted on the panel of yearly AF values
 155 constructed from each LULC measurement series. For each year t and series j , we define

$$AF_{t,j} = \frac{G_t}{FF_t + LULC_{t,j}},$$

156 and estimate

$$AF_{t,j} = (\mu_\alpha + u_{\alpha j}) + (\mu_\beta + u_{\beta j})t + \varepsilon_{t,j},$$

157 with

$$\begin{bmatrix} u_{\alpha j} \\ u_{\beta j} \end{bmatrix} \sim \left(N \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma_u \right), \quad \varepsilon_{t,j} \sim N(0, \sigma_\varepsilon^2).$$

158 This specification allows each LULC series to have its own level and trend while estimating a
 159 common population-average trend μ_β . The main estimand is μ_β , interpreted as the average annual
 160 change in AF across the full set of LULC definitions. Note that this notation is equivalent to the more
 161 compact notation in the main text, where $\alpha_j = \mu_\alpha + u_{\alpha j}$ and $\beta_j = \mu_\beta + u_{\beta j}$.

162 The model is estimated by restricted maximum likelihood^{40,41}. Fixed-effect uncertainty is based on
 163 model-based standard errors and Wald tests. We report the estimated fixed intercept and slope,
 164 standard errors, p-values, and marginal and conditional R-squared^{49,50}.

165 Identification and interpretation rely on standard mixed-model conditions: (i) conditional linearity in
 166 time, (ii) random effects centred at zero and independent across LULC definitions, and (iii) mean-zero
 167 residuals conditional on fixed and random effects.

168 As an endpoint robustness check, we re-estimate the same mixed-effects specification on the sample
 169 ending in 2023 to verify that inference on μ_β is not mechanically driven by the final-year
 170 observation.

171 **Weighted least squares estimation approach**

172 As a robustness check to the mixed-effects model, we also estimate a two-stage measurement-error-
 173 aware WLS model that uses repeated LULC measurements to construct a denominator-uncertainty
 174 proxy for each year and HAC inference^{12–14,44,51}.

175 In the estimation, the yearly denominator is computed as

$$\hat{C}_t = FF_t + \overline{LULC}_t, \quad \overline{LULC}_t = \frac{1}{n_t} \sum_{j=1}^{n_t} LULC_{tj},$$

176 where $n_t = 3$ for the GCB LULC mean and $n_t = 69$ for the all-LULC mean. In the implemented
 177 robustness specification, the yearly weighting term is computed from the cross-measurement
 178 dispersion across the full set of 69 LULC values and is then applied to both denominator definitions.
 179 Assume for each time $t = 1, \dots, T$ we observe:

- 180 • a single numerator G_t (e.g., atmospheric CO_2 growth), and
- 181 • multiple denominator measurements $c_{t1}, \dots, c_{tn_t}$ with $n_t \geq 2$, which are noisy observations of a
 182 latent C_t .

183 Using the repeated denominator measurements, we construct a year-specific uncertainty proxy for
 184 the ratio estimator $AF_t = G_t/C_t$ via the delta method, which captures how dispersion in
 185 denominator definitions propagates into AF. The procedure is described in detail next.

186 **Step 1: Estimate C_t and AF_t**

- 187 1. Construct the denominator estimate at time t as the sample mean across available LULC
 188 measurements as

$$\hat{C}_t = \frac{1}{n_t} \sum_{j=1}^{n_t} c_{tj}.$$

189 We consider two variants of this denominator estimate: one using the GCB LULC column (the mean
 190 of BLUE, OSCAR, LUCE) and one using the all-LULC mean (the mean across all 69 LULC
 191 measurements).

192 The cross-measurement dispersion around $\hat{C}_t$ is estimated from the multiple measurements and used
 193 as a weighting proxy.

194 With $n_t \geq 2$, the empirical dispersion proxy is

$$\widehat{\text{Var}}(\hat{C}_t) = s_{c,t}^2, \quad s_{c,t}^2 = \frac{1}{n_t - 1} \sum_{j=1}^{n_t} (c_{tj} - \bar{c}_t)^2, \quad \bar{c}_t = \hat{C}_t.$$

- 195 2. Form the ratio estimate

$$AF_t = G_t / \hat{C}_t.$$

- 196 3. Use the multivariate delta method (shown below) to obtain an estimate of the AF_t uncertainty

$$\widehat{\text{Var}}(AF_t) = \left(\frac{G_t}{\hat{C}_t^2} \right)^2 s_{c,t}^2.$$

197 **Step 2: WLS regression of AF_t on time**

198 For each time t , we obtain a point estimate AF_t and an uncertainty proxy that reflects cross-
199 measurement dispersion. We use these quantities to fit a linear trend via WLS:

$$AF_t = \alpha + \beta t + \varepsilon_t, \quad \varepsilon_t \sim (0, \sigma_t^2), \quad \sigma_t^2 = \text{Var} (AF_t).$$

200 In our application, the time index t is the year, and AF_t is the estimated AF for that year. The
201 weighting term is obtained from AF uncertainty from denominator dispersion and varies across years
202 with the spread of the LULC measurement panel. We use this term to weight the regression, giving
203 more weight to years with lower denominator-related uncertainty. The core idea is that years with
204 more consistent LULC measurements provide more reliable AF estimates, and hence should have
205 greater influence on the trend estimation.

206 Assuming negligible correlation in ε_t across time, this is WLS with weights $w_t = \frac{1}{\sigma_t^2}$. If we suspect
207 serial correlation, we can use HAC standard errors for inference on β without changing the point
208 estimate. In practice, we use a HAC covariance estimator with a Bartlett kernel and Andrews
209 automatic bandwidth selection¹⁴. In the results, we report both conventional and HAC standard
210 errors.

211 Testing for a time trend is then a test of $\beta = 0$ versus $\beta \neq 0$ in this linear model, and WLS uses the
212 repeated denominator information through this delta-method weighting scheme.

213 **Delta method for ratio variance estimation**

214 **Step-by-step derivation (first-order delta method):**

215 Define the random vector and the function of interest as

$$X_t = \begin{bmatrix} G_t \\ \hat{C}_t \end{bmatrix}, \quad g(X_t) = \frac{G_t}{\hat{C}_t}.$$

216 and let

$$\sigma_{G,t}^2 = \text{Var} (G_t), \quad \sigma_{\hat{C},t}^2 = \text{Var} (\hat{C}_t), \quad \sigma_{GC,t} = \text{Cov} (G_t, \hat{C}_t).$$

217 1. Linearize g around the mean vector $(\mu_{G,t}, \mu_{C,t}) = \mathbb{E}[(G_t, \hat{C}_t)]$:

$$g(X_t) \approx g(\mu_{G,t}, \mu_{C,t}) + \nabla g(\mu_{G,t}, \mu_{C,t})^\top (X_t - \mathbb{E}[X_t]).$$

218 2. Compute the gradient:

$$\nabla g(G, C) = \begin{bmatrix} \partial g / \partial G \\ \partial g / \partial C \end{bmatrix} = \begin{bmatrix} 1/C \\ -G/C^2 \end{bmatrix}.$$

219 3. Write the covariance matrix of $(G_t, \hat{C}_t)$:

$$\Sigma_t = \begin{bmatrix} \sigma_{G,t}^2 & \sigma_{GC,t} \\ \sigma_{GC,t} & \sigma_{C,t}^2 \end{bmatrix}.$$

220 4. Apply the delta-method variance formula

$$\text{Var} (g(X_t)) \approx \nabla g(\mu_{G,t}, \mu_{C,t})^\top \Sigma_t \nabla g(\mu_{G,t}, \mu_{C,t}),$$

221 and with plug-in evaluation at (G_t, C_t) , this becomes

$$\text{Var} (AF_t) \approx \left(\frac{1}{C_t} \right)^2 \sigma_{G,t}^2 + \left(\frac{G_t}{C_t^2} \right)^2 \sigma_{C,t}^2 - 2 \frac{G_t}{C_t^3} \sigma_{GC,t}.$$

222 In practice, we use plug-in estimates (replace unknown moments by their empirical counterparts).

223 This is the standard delta-method approximation for the variance of a ratio estimator.

224 5. In the case analysed in this paper, we have a single numerator measurement from a robust source
 225 per time, so we treat $\sigma_{G,t}^2$ and $\sigma_{GC,t}$ as negligible relative to the variance from the denominator
 226 measurement error, which is captured by $\sigma_{C,t}^2$. Nevertheless, the formula above is general and can
 227 be applied in other contexts where numerator uncertainty is non-negligible or where there is
 228 covariance between numerator and denominator measurements.

229 Under the assumption of negligible numerator variance and covariance, the formula simplifies to

$$\text{Var} (AF_t) \approx \left(\frac{G_t}{C_t^2} \right)^2 \sigma_{C,t}^2.$$

230 Accordingly, a practical plug-in variance estimator for AF_t is

$$\widehat{\text{Var}}(AF_t) = \left(\frac{G_t}{C_t^2} \right)^2 s_{c,t}^2.$$

231 **Data availability**

232 The study uses publicly available Global Carbon Budget 2025 data and derived yearly series
 233 generated from those source files. The processed analysis tables are provided in the manuscript
 234 repository under the results directory, and the extracted and derived LULC panel is available as a
 235 CSV file in the data directory.

236 **Code availability**

237 All code used to process data, estimate models, and generate figures is available in the repository
238 (github.com/everval/Airborne-Fraction-WLS-Trend) under the scripts directory, including Quarto
239 analysis files and Julia helper functions.

240 **Competing interests**

241 The author declares no competing interests.

242 **Additional information**

243 Supplementary information is available in the accompanying supplementary manuscript file.

244 **Bibliography**

- 245 1. Canadell, J. G. *et al.* Contributions to accelerating atmospheric CO₂ growth from economic
246 activity, carbon intensity, and efficiency of natural sinks. *Proceedings of the National Academy of*
247 *Sciences* **104**, 18866–18870 (2007).
- 248 2. Raupach, M. R. *et al.* Global and regional drivers of accelerating CO₂ emissions. *Proceedings of*
249 *the National Academy of Sciences* **104**, 10288–10293 (2007).
- 250 3. Knorr, W. Is the airborne fraction of anthropogenic CO₂ emissions increasing?. *Geophysical*
251 *Research Letters* **36**, L21710 (2009).
- 252 4. Ballantyne, A. P., Alden, C. B., Miller, J. B., Tans, P. P. & White, J. W. C. Increase in observed net
253 carbon dioxide uptake by land and oceans during the past 50 years. *Nature* **488**, 70–72 (2012).
- 254 5. Le Quéré, C. *et al.* Trends in the sources and sinks of carbon dioxide. *Nature Geoscience* **2**, 831–
255 836 (2009).
- 256 6. Bennedsen, M., Hillebrand, E. & Koopman, S. J. On the evidence of a trend in the CO₂ airborne
257 fraction. *Nature* **616**, E1–E3 (2023).
- 258 7. Bennedsen, M., Hillebrand, E. & Koopman, S. J. A regression-based approach to the CO₂ airborne
259 fraction. *Nature Communications* **15**, 8507 (2024).
- 260 8. Friedlingstein, P. *et al.* Global carbon budget 2025. *Earth System Science Data* **18**, 3211–3288
261 (2026).
- 262 9. Hansis, E., Davis, S. J. & Pongratz, J. Relevance of methodological choices for accounting of land
263 use change carbon fluxes. *Global Biogeochemical Cycles* **29**, 1230–1246 (2015).
- 264 10. Gasser, T. *et al.* Historical CO₂ emissions from land use and land cover change and their
265 uncertainty. *Biogeosciences* **17**, 4075–4101 (2020).

- 266 11. Qin, Z. *et al.* Global spatially explicit carbon emissions from land-use change over the past six
267 decades (1961-2020). *One Earth* **7**, 835–847 (2024).
- 268 12. Fuller, W. A. *Measurement Error Models*. (Wiley, New York, 1987).
- 269 13. Carroll, R. J., Ruppert, D., Stefanski, L. A. & Crainiceanu, C. M. *Measurement Error in Nonlinear*
270 *Models: A Modern Perspective*. (Chapman & Hall/CRC, Boca Raton, 2006).
- 271 14. Newey, W. K. & West, K. D. A Simple, Positive Semi-definite, Heteroskedasticity and
272 Autocorrelation Consistent Covariance Matrix. *Econometrica* **55**, 703–708 (1987).
- 273 15. Lan, X., Tans, P. & Thoning, K. W. Trends in globally-averaged CO₂ determined from NOAA
274 Global Monitoring Laboratory measurements. <https://gml.noaa.gov/ccgg/trends/global.html>
275 (2026) doi:10.15138/9N0H-ZH07.
- 276 16. Conchedda, G. & Tubiello, F. N. Drainage of organic soils and GHG emissions: validation with
277 country data. *Earth System Science Data* **12**, 3113–3137 (2020).
- 278 17. Müller, J. & Joos, F. Committed and projected future changes in global peatlands – continued
279 transient model simulations since the Last Glacial Maximum. *Biogeosciences* **18**, 3657–3687
280 (2021).
- 281 18. Qiu, C. *et al.* Large historical carbon emissions from cultivated northern peatlands. *Science*
282 *Advances* **7**, eabf1332 (2021).
- 283 19. Haverd, V. *et al.* A new version of the CABLE land surface model (Subversion revision r4601)
284 incorporating land use and land cover change, woody vegetation demography, and a novel
285 optimisation-based approach to plant coordination of photosynthesis. *Geoscientific Model*
286 *Development* **11**, 2995–3026 (2018).
- 287 20. Melton, J. R. *et al.* CLASSIC v1.0: the open-source community successor to the Canadian Land
288 Surface Scheme (CLASS) and the Canadian Terrestrial Ecosystem Model (CTEM) – Part 1: Model
289 framework and site-level performance. *Geoscientific Model Development* **13**, 2825–2850 (2020).
- 290 21. Lawrence, D. M. *et al.* The Community Land Model version 5: Description of new features,
291 benchmarking, and impact of forcing uncertainty. *Journal of Advances in Modeling Earth Systems*
292 **11**, 4245–4287 (2019).
- 293 22. Fisher, R. A. *et al.* Taking off the training wheels: the properties of a dynamic vegetation model
294 without climate envelopes, CLM4.5(ED). *Geoscientific Model Development* **8**, 3593–3619 (2015).
- 295 23. Tian, H. *et al.* North American terrestrial CO₂ uptake largely offset by CH₄ and N₂O emissions:
296 Toward a full accounting of the greenhouse gas budget. *Climatic Change* **129**, 423–426 (2015).

- 297 24. Ma, L. *et al.* Global evaluation of the Ecosystem Demography model (ED v3.0). *Geoscientific*
298 *Model Development* **15**, 1971–1994 (2022).
- 299 25. Yang, X., Thornton, P., Ricciuto, D., Wang, Y. & Hoffman, F. Global evaluation of terrestrial
300 biogeochemistry in the Energy Exascale Earth System Model (E3SM) and the role of the
301 phosphorus cycle in the historical terrestrial carbon balance. *Biogeosciences* **20**, 2813–2836 (2023).
- 302 26. Needham, J. & others. Vertical canopy gradients of respiration drive plant carbon budgets and
303 leaf area index. *New Phytologist* **246**, 144–157 (2025).
- 304 27. Felzer, B. S. & Jiang, M. Effect of land use and land cover change in context of growth
305 enhancements in the United States since 1700: Net source or sink?. *Journal of Geophysical*
306 *Research: Biogeosciences* **123**, 3439–3457 (2018).
- 307 28. Xia, J. Z. *et al.* The carbon budget of China: 1980–2021. *Science Bulletin* **69**, 114–124 (2024).
- 308 29. Yue, X. *et al.* Development and evaluation of the interactive Model for Air Pollution and Land
309 Ecosystems (iMAPLE) version 1.0. *Geoscientific Model Development* **17**, 4621–4642 (2024).
- 310 30. Shu, S., Jain, A. K., Koven, C. D. & Mishra, U. Estimation of Permafrost SOC Stock and Turnover
311 Time Using a Land Surface Model With Vertical Heterogeneity of Permafrost Soils. *Global*
312 *Biogeochemical Cycles* **34**, e2020GB006585 (2020).
- 313 31. Reick, C. H. *et al.* JSBACH 3 - The land component of the MPI Earth System Model:
314 documentation of version 3.2. (2021) doi:10.17617/2.3279802.
- 315 32. Poulter, B., Frank, D. C., Hodson, E. L. & Zimmermann, N. E. Impacts of land cover and climate
316 data selection on understanding terrestrial carbon dynamics and the CO₂ airborne fraction.
317 *Biogeosciences* **8**, 2027–2036 (2011).
- 318 33. Smith, B. *et al.* Implications of incorporating N cycling and N limitations on primary production
319 in an individual-based dynamic vegetation model. *Biogeosciences* **11**, 2027–2054 (2014).
- 320 34. Schaphoff, S. *et al.* LPJmL4 – a dynamic global vegetation model with managed land – Part 1:
321 Model description. *Geoscientific Model Development* **11**, 1343–1375 (2018).
- 322 35. Lienert, S. & Joos, F. A Bayesian ensemble data assimilation to constrain model parameters and
323 land-use carbon emissions. *Biogeosciences* **15**, 2909–2930 (2018).
- 324 36. Vuichard, N. *et al.* Accounting for carbon and nitrogen interactions in the global terrestrial
325 ecosystem model ORCHIDEE (trunk version, rev 4999): multi-scale evaluation of gross primary
326 production. *Geoscientific Model Development* **12**, 4751–4779 (2019).

37. Walker, A. P. *et al.* The impact of alternative trait-scaling hypotheses for the maximum photosynthetic carboxylation rate (V-cmax) on global gross primary production. *New Phytologist* **215**, 1370–1386 (2017).
38. Kato, E., Kinoshita, T., Ito, A., Kawamiya, M. & Yamagata, Y. Evaluation of spatially explicit emission scenario of land-use change and biomass burning using a process-based biogeochemical model. *Journal of Land Use Science* **8**, 104–122 (2013).
39. Ito, A. Disequilibrium of terrestrial ecosystem CO₂ budget caused by disturbance-induced emissions and non-CO₂ carbon export flows: a global model assessment. *Earth System Dynamics* **10**, 685–709 (2019).
40. Henderson, C. R. Estimation of Variance and Covariance Components. *Biometrics* **9**, 226–252 (1953).
41. Bolker, B. M. *et al.* Generalized linear mixed models: a practical guide for ecology and evolution. *Trends in Ecology & Evolution* **24**, 127–135 (2009).
42. Gloor, M. & others. What Can Be Learned about Carbon Cycle Climate Feedbacks from the CO₂ Airborne Fraction?. *Atmospheric Chemistry and Physics* **10**, 7739–7752 (2010).
43. Vera-Valdés, J. E. & Grivas, C. Robust estimation of carbon dioxide airborne fraction under measurement errors. *Environmental Research Communications* **7**, 31009 (2025).
44. Oehlert, G. W. A Note on the Delta Method. *The American Statistician* **46**, 27–29 (1992).
45. Minière, A., Schuckmann, K. von, Sallée, J.-B. & Vogt, L. Robust acceleration of Earth system heating observed over the past six decades. *Scientific Reports* **13**, 22975 (2023).
46. World Meteorological Organization. State of the Global Climate 2025. <https://wmo.int/publication-series/state-of-global-climate/state-of-global-climate-2025> (2026) doi:10.59327/WMO/S/CRI/SOC/1.
47. Storto, A. & Yang, C. Acceleration of the ocean warming from 1961 to 2022 unveiled by large-ensemble reanalyses. *Nature Communications* **15**, 545 (2024).
48. Foster, G. & Rahmstorf, S. Global Warming Has Accelerated Significantly. *Geophysical Research Letters* **53**, e2025GL118804 (2026).
49. Nakagawa, S. & Schielzeth, H. A General and Simple Method for Obtaining R^2 from Generalized Linear Mixed-Effects Models. *Methods in Ecology and Evolution* **4**, 133–142 (2013).

- 356 50. Nakagawa, S., Johnson, P. C. D. & Schielzeth, H. The Coefficient of Determination R^2 and
357 Intra-Class Correlation Coefficient from Generalized Linear Mixed-Effects Models Revisited and
358 Expanded. *Journal of the Royal Society Interface* **14**, 20170213 (2017).
- 359 51. Aitken, A. C. On Least Squares and Linear Combination of Observations. *Proceedings of the Royal*
360 *Society of Edinburgh* **55**, 42–48 (1935).